

Music-Gen v4 closure campaign — cycles 10-12

2026-09-04

Contents

Music-Gen v4 closure campaign — cycles 10-12	1
Abstract	1
1. Where things stood at cycle-10 open	1
2. Cycle 10 — disk-check repair and drums sweep launch	2
3. Cycle 11 — drums fine-fit, channel-aware replay, family verdict	2
4. Cycle 12 — family-2 arc, arc closeout, and independent replay regression	4
5. Findings closed across the batch	7
6. Systematic finding across cycles 1-12	7
7. Audit-cycle trajectory	8
8. Cycle-13 scope	8
Appendix: Implementation Details	9

Music-Gen v4 closure campaign — cycles 10-12

Abstract

Cycles 10-12 close the Chicken Grease drums sub-milestone of the v4 closure campaign. Cycle 10 fixed the cycle-9 disk-check false positive that had blocked the detached SoundFont drums sweep and launched the sweep under the canonical replay-time environment pin. Cycle 11 ran the 216-cell fine-fit stage, emitted the drums profile of record (program 16 Power Kit, embedding-cosine 0.2374 against the reference stem), and patched the replay dispatcher to be MIDI-channel aware so that channel-10 drum programs replay correctly; the SoundFont family verdict came out **SF2_RULED_OUT** because the top-1 embedding cosine sits below the 0.40 retained honesty floor. Cycle 12 shipped the family-2 stem-sampled drums arc — spike, builder, render, replay proof, verdict — and ruled it out as well at embedding cosine 0.0372, closing the drums arc as **CG_DRUMS_ARC_EXHAUSTED_NO_CONFIRMED** with the same shape as the cycle-7 bass closeout. It also ran an independent from-fresh-subprocess replay-regression check covering both the bass-v2 anchor and the new drums anchor (both byte-identical across two runs), closing the moderate finding the cycle-11 auditor had left open. A three-option drums acceptance policy has been escalated to operator authority; unlike the cycle-9 bass case, the agent did not unilaterally extend the composite-relative acceptance precedent, because the operator's cycle-9 threshold retirement was explicitly scoped to Chicken Grease bass. Two of the five Chicken Grease instrument tracks now have terminal verdicts; the full A/B render remains gated on the remaining three sub-milestones plus the drums acceptance decision.

1. Where things stood at cycle-10 open

By the close of cycle 9, three loose ends remained under M-V4-PROFILES-1:

1. The detached SoundFont drums coarse sweep had been prepared (script `coarse_sweep_sf2_drums.py`, dry-run PASS) but its launch had been halted by a false positive in the pre-flight disk-space check. The check used a `statvfs`-derived percentage that reported the

working volume at 97.39 % full, while `df -h` on the same volume showed 82.24 % full and 6.6 GB free — comfortably above the 500 MB working-audio budget of a single sweep cycle. This was the sole gate on running the drums sweep.

2. The reference render family (SoundFont) had been proved replayable for bass-only material (MIDI channel 0), but the replay engine had not yet been exercised on a channel-10 drum-kit program.
3. The Chicken Grease A/B render scaffold (`deliver_cg_ab_v4.py`) had smoke-tested with `n_missing = 4` — drums, piano, guitar, and the residual “other” bucket — and was waiting for each of those profiles to land.

The cycle-10 auditor also carried forward a broader observation from prior cycles: percentage-based resource-limit checks were prone to mismatching the reference tool’s numeric intuition (`df` vs `statvfs`, LUFs-I vs RMS-dBFS fallback, `sha16` slicing), and a small shared `long_exposure.tools.resource_check` helper module would prevent the class of defect recurring. That refactor was noted and deferred; the concrete cycle-10 work was the point fix.

2. Cycle 10 — disk-check repair and drums sweep launch

The `_disk_ok()` predicate in `coarse_sweep_sf2_drums.py` was rewritten from a percentage-remaining formula to an absolute-budget formula:

$$\text{ok} \Leftrightarrow \text{avail_bytes} \geq \text{budget_bytes} \cdot \text{safety_factor}$$

with `budget_bytes = 500 · 220` and `safety_factor = 1.5`. This matches the operator’s stated budget discipline directly (a working sweep may consume up to 500 MB of audio) rather than backing it out from a percentage of a volume whose size varies with mount point. A regression test, `tests/test_coarse_sweep_disk_check.py`, was added that pins the new formula and asserts the cycle-9 false-positive scenario (large volume, small percentage-free, ample absolute-free) now passes the check.

With the gate repaired, the detached drums sweep launched under the canonical seven-key replay-time environment pin (`LC_ALL, MKL_NUM_THREADS, OMP_NUM_THREADS, OPENBLAS_NUM_THREADS, PYTHONHASHSEED, SOURCE_DATE_EPOCH, TZ`; SHA `2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842a922ca`). The coarse sweep enumerated the 15 GM drum-kit programs on MIDI channel 10 against the reference stem (`stems_6s/drums.wav`, SHA `34492c03f301b6eac3a75343b61244193889d039ae4ccce4c35cc44d568ac835`) and the drums MIDI excerpt (channel 10, 186 note-on events, SHA `0fd71ce70a26365c2acf08b9f87531178f9f9c18cc4`). The stage-1 leaderboard landed at `drums_sweep_stage1/leaderboard.tsv` with SHA `dd5544d3bd3a549cab95...`. Two supplementary regression tests covering the drums-plus-bass cross-check and the drums-family stage-2 leaderboard structure — `test_rc10_drums_bass.py` and `test_rc10_drums_v2.py` — were added.

The cycle-10 auditor raised one moderate finding: the replay engine’s `_replay_sf2` had never been exercised on a program on a MIDI channel other than 0, and the bass-only regression that had validated the cycle-6 replay fix therefore did not cover the drums case. That finding was scheduled for cycle-11 closure.

3. Cycle 11 — drums fine-fit, channel-aware replay, family verdict

3.1 Stage-2 fine-fit and profile emission

A drums-specific fine-fit driver, `fine_fit_sf2_drums.py` (679 lines, channel-10 aware, LUFs-normalised) was authored as a direct sibling of the cycle-3 bass driver. It re-scored the top-3

coarse programs across a 216-cell grid over gain, reverb-send, and post-processing. The stage-2 leaderboard (drums_sweep_stage2/leaderboard.tsv, SHA 81a441732f7f7d1da615...) shows a composite spread of 176.7 % across the 216 cells, indicating that the fine-fit is not degenerate.

The stage-2 top-1 by frozen composite (weights 0.5 mel-L1, 0.25 spectral-centroid RMSE, 0.25 embedding-cosine) came in at:

field	value
GM program	16 (Power Kit)
gain	1.0
reverb send	0.7
post-processing	EQ_only
composite	475.74
mel-L1	10.44 dB
spectral-centroid RMSE	1858.33 Hz
embedding-cosine (VGGish)	0.2374

The drums profile was pinned as drums.json, UUID 83728154-6f48-5c5d-a558-b4d82523ac1b, canonical replay SHA dadafcfc0153f002651c23975c3845dd3f8ca7896d263faf1c52eb54d64b8d7c.

3.2 Channel-aware replay fix

To close the cycle-10 moderate finding, replay.py was patched in place at lines 79–93. The pre-patch dispatcher wrote the program-change event onto channel 0 regardless of the program’s actual GM channel; for a channel-10 drum kit this either silently mis-routes to a melodic channel or drops the event depending on the SoundFont’s channel mapping. The patched dispatcher reads the target channel from the profile identity, rewrites the MIDI to strip any inbound program-changes, and injects a program_change(channel=<channel>, program=<program>, time=0) at tick 0.

The regression discipline for a replay-engine change requires that every prior anchor whose MIDI is exercised by the new code path reproduces byte-identical. Because the bass MIDI is channel-0-only, the _replay_sf2 extension collapses to the pre-patch behaviour on that input; the bass-v2 anchor 832868d0ea8a81cab2569e60445f80d516d1b5bb958b1b8b0c2e996bdb3aeac5 is reproduced exactly. The drums anchor is a new render on a new code path; it is captured in the drums replay proof (drums.replay_proof.json) with run1 = run2 = dadafcfc0153f002651c23975c3845dd3f8ca7896d263faf1c52eb54d64b8d7c under the canonical seven-key environment pin. The proof file carries a bass_regression_check block that pins the bass-v2 byte-identity outcome explicitly.

Post-patch replay.py SHA is 1f43027039c45f5e066c.... This file becomes a read-only anchor from cycle 11 forward; any subsequent change to the replay engine must pass a regression check against both the bass-v2 and drums anchors (and, from cycle 12, the family-2 drums anchor as well).

3.3 Family verdict

The stage-2 leaderboard was scored against the pre-registered decision protocol. The CONFIRMED gate (embedding-cosine ≥ 0.60 and stage-2 spread $\geq 10\%$ of the cycle-10 coarse spread) is met on the spread clause (176.7 % ≥ 8.97 %) but not on the embedding clause

(0.2374 < 0.60). The retained absolute RULED_OUT floor (embedding-cosine $\leq$ 0.40) is satisfied (0.2374 $\leq$ 0.40). Verdict: **SF2_RULED_OUT**.

drums_family_verdict.json records the outcome as a first-class negative finding rather than a failure. In particular it discloses the honesty context that would otherwise be hidden by a composite-only summary:

honesty disclosure	value
top-1 embedding-cosine (Power Kit, prog 16)	0.2374
max embedding-cosine across all 216 stage-2 cells	0.4645 (program 48 Orchestra Kit, composite rank 76)
program 0 Standard Kit best/worst rank	13 / 216
distinct render SHAs across 216 cells	216 / 216
EQ v2 zero-mean removal effective	yes

The verdict pins the decision-protocol scope: the composite-relative WINNER extension that the operator authorised for Chicken Grease bass at cycle 9 is explicitly scoped to STILL_INDETERMINATE outcomes and does **not** rescue a RULED_OUT — the retained 0.40 floor is a hard honesty gate. Any attempt to accept the Power Kit as-is would require an explicit operator scope extension.

The verdict also names a parallel to the cycle-1 bass surprise (where the frozen composite preferred a Drawbar Organ over any bass program): the same objective on Chicken Grease drums prefers the Power Kit over the Orchestra Kit and both over the Standard Kit, none of which reach the CONFIRMED gate. The pattern is recorded as a systematic characterisation of the objective on Chicken Grease content, not a defect.

The cycle-11 auditor raised one moderate finding: the channel-aware replay fix had been reproduced by the worker but had not yet been re-run from an independent Python subprocess under the same environment pins — the strongest form of the regression check. That finding was scheduled for cycle-12 closure.

4. Cycle 12 — family-2 arc, arc closeout, and independent replay regression

Cycle 12 ran three concurrent tracks that closed the drums arc cleanly.

4.1 Independent replay-regression verification

A stand-alone harness _replay_regression_c12.py was written to close the cycle-11 moderate. It forks a fresh Python subprocess for each anchor (so it inherits none of the parent’s imported state), sets the seven-key canonical environment pin, replays the target profile into a fresh temporary directory, computes the SHA of the resulting WAV, and repeats the entire flow a second time into a second temporary directory. The harness targets both the bass-v2 anchor and the new drums anchor.

The verdict (_replay_regression_c12.json) is **REPLAY_REGRESSION_HOLDS**. Both anchors reproduce byte-identical across two fresh-subprocess runs and match the anchor SHAs on file:

label	run 1 SHA	run 2 SHA	anchor match
bass_v2	832868d0...aeac5	832868d0...aeac5	☐

label	run 1 SHA	run 2 SHA	anchor match
drums	dadafcfc...64b8d7c	dadafcfc...64b8d7c	☐

The harness also caught a transcription error in its own inbound research brief: the brief had pinned the drums anchor with a wrong tail (first 16 hex characters `dadafcfc0153f002` correct, remainder divergent). The harness cited the on-disk canonical value from `drums.replay_proof.json`, disclosed both variants in a `brief_anchor_discrepancy_note` block, and reproduced the on-disk value byte-identical rather than retro-fitting to the incorrect value. The cycle-12 auditor re-ran the same harness independently from its own fresh subprocess and reproduced both anchor SHAs, validating the closure.

4.2 Family-2 stem-sampled drums arc

The family-2 shape probe (`family2_stem_sampled_drums_spike.py`, 144 lines) detected 147 onsets in the reference drums stem over a 6.9 s window and classified them by a band-energy argmax into kick / snare / hi-hat buckets. The spike verdict was `VARIABLE`, but the classifier produced an asymmetric class distribution (kick 93, snare 0, hi-hat 53) versus the MIDI-side pitch histogram (kick 30, snare 33, hi-hat 123). This asymmetry was surfaced in the spike JSON rather than smoothed over.

The builder (`family2_stem_sampled_drums_builder.py`, 271 lines) rendered the drums MIDI concatenatively by MIDI-pitch → sample-class dispatch (pitch 36 → kick sample, pitches 37–38 → snare, pitches 42/44/46 → hi-hat) using the sample bank that the spike had extracted from the reference stem. The render is `drums_family2_render/render.wav`, 2 678 664 bytes, SHA `69a76c5b4498972d1cb878da94e645c8c341675b113cc4ca315435f6bb16ca00`.

Because family-2 is a distinct render family from SoundFont under the file-determinism policy FD-16(c), it requires its own replay proof. `drums_family2.replay_proof.json` records `run1 = run2 = 69a76c5b...16ca00` under the canonical seven-key pin. This single proof covers the family-2 code path for all future stem-sampled CG drums profiles.

The family-2 profile of record is `drums_family2_v1.json`, UUID `13ae0000-934e-5b4c-9a7a-e69e1c0e5fc4`. Scored against the frozen composite:

field	value
composite	618.16
mel-L1	13.41 dB
spectral-centroid RMSE	2442.08 Hz
embedding-cosine (VGGish)	0.0372

The verdict is **FAMILY2 RULED_OUT**: embedding-cosine 0.0372 is well below the 0.40 retained honesty floor. This is the second family-2 verdict on Chicken Grease content and is even lower than the cycle-6 family-2 bass score of 0.0896.

A minor observation was recorded for future revisit: the family-2 profile’s `params.classifier` field names the band-energy spike-side classifier used to extract the sample bank, but the builder’s render-time dispatch is by MIDI pitch class rather than by band energy. Both routings are internally consistent, but the field name is imprecise; a companion `params.render_dispatch` field would disambiguate. Non-blocking because the arc closes on `RULED_OUT`.

4.3 Arc closeout and manager escalation

With both frozen render families ruled out, the drums arc is formally closed in `drums_arc_closeout.json` under the verdict **CG_DRUMS_ARC_EXHAUSTED_NO_CONFIRMED**. The closeout parallels the cycle-7 bass arc closeout in shape:

dimension	cg-bass (c7)	cg-drums (c12)
SoundFont top-1 embedding-cosine	0.4946 (prog 33 Electric Bass Finger)	0.2374 (prog 16 Power Kit)
SoundFont max embedding-cosine across 216	0.4946 (prog 19 Church Organ)	0.4645 (prog 48 Orchestra Kit, composite rank 76)
SoundFont verdict family-2 embedding-cosine	STILL_INDETERMINATE 0.0896	SF2_RULED_OUT 0.0372
family-2 verdict resolution	FAMILY2_RULED_OUT operator OPT1+OPT3 hybrid at c9	FAMILY2_RULED_OUT manager escalation to operator; no unilateral acceptance

The closeout carries a `cross_song_parallel_findings` block naming the bass precedent and a `systematic_finding` block characterising the pattern: the concatenative stem-sampled render approach as currently implemented (fixed windows, per-class sample bank without pitch modelling) is architecturally insufficient to reach the 0.40 floor on Chicken Grease content across two independent instrument families. If the pattern holds, piano, guitar, and other-residual arcs are likely to exhaust similarly and require comparable operator adjudication.

The drums acceptance policy is escalated to operator authority in `_manager_M-V4-SHOWCASE-1-cg-drums-acceptance-policy.json` with three named options:

option	label	consequence
OPT1	Accept SoundFont top-1 as drums WINNER via composite-relative extension	requires operator threshold-retirement scope extension; the cycle-9 retirement is scoped to bass only
OPT2	Accept SoundFont max-embedding-cosine candidate (embedding-first tiebreak)	small cost: single fine-fit lookup + one render + one replay proof for program 48 Orchestra Kit (embedding-cosine 0.4645); does not require composite-relative extension
OPT3	Refuse drums showcase; deliver Chicken Grease A/B without drums recreation	delivers with four instruments (bass_v2 + piano + guitar + other-residual, once profiled) using original htdemucs drums stem

Unlike cycle 9 — where the agent had unilaterally chosen to accept the composite-relative

WINNER for bass after receiving the operator directive — the cycle-12 escalation explicitly declines to pre-empt. The escalation JSON pins `unilateral_action_taken_this_cycle = "NONE"` and instructs the next cycle to wait on operator input in `live_guidance`. The reason is that the cycle-9 operator directive scoped the threshold retirement to Chicken Grease bass by name; extending it to drums is a scope decision, not a mechanical follow-through.

4.4 Ledger discipline

Nine ledger events were emitted at cycle 12 in strict timestamp order (00:20 UTC through 00:51 UTC): the family-2 profile, replay proof, verdict, arc closeout, manager escalation, replay regression, the plan-of-record registration (`_plan/register-c12-cg-drums-family2-sub-leaves`), and two housekeeping entries (`_archive/cycle-12-scratch`, `_infra/adopt-cycle12-tests`). Every event carries a UUID5 identifier and the manager escalation is emitted with status: `action_required` as required by the state machine.

5. Findings closed across the batch

finding	opened	closed	resolution
disk-check false positive halts drums sweep	c9	c10	<code>_disk_ok()</code> rewritten to absolute-budget formula; regression test pins the c9 scenario
replay engine not exercised on non-channel-0 program	c10	c11	<code>replay.py</code> L79-93 channel-aware fix; drums renders replayable; bass-v2 byte-identity preserved
channel-aware fix not independently re-verified from fresh subprocess	c11	c12	<code>_replay_regression_ c12.py</code> reproduces bass-v2 and drums anchors byte-identical from fresh subprocesses; auditor independently re-runs and reproduces both

The cycle-12 auditor's own severity tally was **CRITICAL 0, MODERATE 0, MINOR 2**. The two minor observations — the spike-vs-builder classifier-nomenclature drift and the zero-snare asymmetry in the band-energy classifier — are recorded in the cycle-12 verdict artefacts and do not gate the arc close.

6. Systematic finding across cycles 1-12

Two consecutive Chicken Grease instrument arcs have exhausted the two frozen render families without reaching the 0.60 CONFIRMED gate:

instrument	SoundFont top-1 emb-cos	family-2 emb-cos	resolution
bass	0.4946	0.0896	operator OPT1+OPT3 hybrid (c9)
drums	0.2374	0.0372	escalation to operator, three named options (c12)

Family-2 stem-sampled underperforms family-1 SoundFont on Chicken Grease content by roughly an order of magnitude in embedding cosine on both instruments. The cycle-12 arc closeout records this as a characterisation of the objective and the render families on this piece of source content, not as a defect of any single component. The convergent expectation is that piano, guitar, and other-residual will follow the same pattern and each require operator adjudication of an acceptance policy; each such adjudication is honest and non-idle, and the campaign converges even under the pattern.

7. Audit-cycle trajectory

The three audits landed cleanly in a convergent trajectory:

audit	severity summary	closure
cycle-10	1 MODERATE (channel-aware replay coverage gap)	closed cycle 11
cycle-11	1 MODERATE (independent from-fresh-subprocess re-verify)	closed cycle 12
cycle-12	0 MODERATE, 2 MINOR	non-blocking; carried as observations

The cycle-12 auditor explicitly re-ran the cycle-12 replay-regression harness from an independent Python subprocess under the canonical seven-key pin and reproduced both anchor SHAs byte-identical, which is the strongest form of the Track-1 verification: the worker's report is validated by execution, not by re-reading their claim.

8. Cycle-13 scope

1. Register the drums acceptance-policy outcome. If an operator directive arrives, apply it verbatim and pin the resulting drums delivery profile. If none arrives, the cycle-9 banned- heartbeat rule instructs the agent to pick the option most consistent with the binding specifications and best objective evidence, disclose the acceptance-fork in the plan of record verbatim, and proceed; the cycle-12 auditor guidance names OPT1 as consistent with the cycle-9 pattern and OPT2 as the embedding-first alternative that does not require scope extension.
2. Open the Chicken Grease piano SoundFont coarse sweep as a direct sibling of `coarse_sweep_sf2.py` and `coarse_sweep_sf2_drums.py`. GM programs 0–7 and 16–19, MIDI channel 1, standard 500 MB working-audio budget and `--score-and-delete --keep-top 3` hygiene.

3. Back-fill accumulated test debt for the family-2 drums concatenative render path — a dedicated `tests/test_sound_match_family2_drums.py` pinning the family-2 render SHA 69a76c5b... would close three cycles of honestly-deferred test coverage cheaply.

Read-only anchors that must be preserved: `M-V4-CERT-1`, `bass.json`, `bass_v2.json`, both bass family verdicts, the bass arc closeout, `drums.json`, `drums.replay_proof.json`, `drums_family_verdict.json`, `drums_family2_v1.json`, `drums_family2.replay_proof.json`, `drums_family2_verdict.json`, `drums_arc_closeout.json`, and the frozen script set in `scripts/sound_match/` (`coarse_sweep_sf2*.py`, `fine_fit_sf2*.py`, `family2_stem_sampled*.py`, `replay.py`). The 0.60 CONFIRMED and 0.40 RULED_OUT thresholds remain in force globally; the cycle-9 threshold retirement is scoped to Chicken Grease bass only.

Appendix: Implementation Details

A.1 Files created or extended, cycles 10-12

Scripts (*scripts/sound_match/*):

file	cycle	lines	role
<code>coarse_sweep_sf2_drums.py</code>	patched 10	446	<code>_disk_ok()</code> rewritten to absolute-budget check
<code>_launch_cg_drums_sweep_c10.sh</code>	10	—	detached launcher for the drums coarse sweep
<code>fine_fit_sf2_drums.py</code>	11	679	216-cell drums stage-2 fine-fit, channel-10 aware
<code>_launch_cg_drums_stage2_c11.sh</code>	11	—	detached launcher for the drums stage-2 fine-fit
<code>replay.py</code>	patched 11	166	channel-aware <code>_replay_sf2</code> extension; post-patch SHA 1f43027039c45f5e066c...
<code>family2_stem_sampled_drums_spike.py</code>	12	144	family-2 shape probe, band-energy classifier
<code>family2_stem_sampled_drums_builder.py</code>	12	271	family-2 concatenative render, MIDI-pitch dispatch
<code>_family2_drums_score_and_emit_c12.py</code>	12	—	scoring + profile + replay-proof + verdict emitter
<code>_family2_drums_closeout_and_escalation_c12.py</code>	12	—	arc closeout + manager escalation emitter
<code>_replay_regression_c12.py</code>	12	—	independent from-fresh-subprocess
<code>_emit_c12_ledger_events.py</code>	12	—	Track-1 harness cycle-12 ledger-event emitter

Tests (*tests/*):

file	cycle	role
test_coarse_sweep_disk_check.py	10	pins the absolute-budget formula and asserts the c9 false-positive scenario
test_rc10_drums_bass.py	10	drums-plus-bass cross-check under channel-aware replay
test_rc10_drums_v2.py	10	drums-family stage-2
test_verdict_sha_fields_resolve_on_disk.py	12	leaderboard structural test asserts every SHA field in a verdict/manifest resolves byte-identical on disk

A.2 Data artefacts, cycles 10-12

All under data/v4/profiles/31a164f845f8e27e/ unless noted.

file	cycle	notes
drums_sweep_stage1/leaderboard.tsv	10	15-preset SoundFont drums coarse-sweep leaderboard, SHA dd5544d3bd3a549cab95...
drums_sweep_stage1/drums_excerpt.mid	10	channel-10 drums MIDI, 186 note-on events, SHA 0fd71ce70a26365c2acf...
drums_sweep_stage2/leaderboard.tsv	11	216-cell drums stage-2 leaderboard, SHA 81a441732f7f7d1da615..., composite spread 176.7 %
drums.json	11	drums profile, UUID 83728154-6f48-5c5d-a558-b4d82523ac1b, program 16 Power Kit, embedding-cosine 0.2374
drums.replay_proof.json	11	run1 = run2 = dadafcfc...64b8d7c; embeds bass-v2 regression PASS
drums_family_verdict.json	11	SF2 RULED OUT ; disclosure block names max-emb-cos 0.4645 at prog 48 rank 76
drums_family2_spike_c12.json	12	shape-probe verdict VIABLE; 147 onsets, kick 93 / snare 0 / hihat 53
drums_family2_v1.json	12	family-2 profile, UUID 13ae0000-934e-5b4c-9a7a-e69e1c0e5fc4, canonical replay SHA 69a76c5b...16ca00
drums_family2.replay_proof.json	12	run1 = run2 = 69a76c5b...16ca00; covers family-2 code path per FD-16(c)

file	cycle	notes
drums_family2_ render/render.wav	12	2 678 664 B family-2 concatenative render
drums_family2_verdict.json	12	FAMILY2_RULED_OUT , embedding-cosine 0.0372
drums_arc_closeout.json	12	CG_DRUMS_ARC_EXHAUSTED_NO_ CONFIRMED ; parallels cycle-7 bass closeout shape
_manager_M-V4-SHOWCASE-1-cg- drums-acceptance-policy.json	12	three named options (OPT1/OPT2/OPT3); authority = OPERATOR; unilateral_action_taken_this_ cycle = NONE
_replay_regression_c12.json	12	REPLAY_REGRESSION_HOLDS ; bass-v2 and drums both byte-identical, discloses brief-vs-on-disk tail discrepancy
_c12_track3_summary.json, _c12_track4_summary.json	12	per-track sub-milestone summaries

A.3 Environment pins in force

The canonical seven-key replay-time pin 2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842a922ca covers every cycle-10-12 render, replay proof, verdict, and regression check without change. It comprises: LC_ALL=C.UTF-8, MKL_NUM_THREADS=1, OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, PYTHONHASHSEED=0, SOURCE_DATE_EPOCH=1756463424, TZ=UTC. The cycle-11 replay.py patch was verified to leave the pin unchanged, and the cycle-12 independent regression re-verified pin, replay engine, and anchor SHAs jointly.

A.4 Read-only anchors preserved

Every prior anchor exercised by cycles 10-12 was re-checked at audit time and found byte-identical. Highlights:

anchor	SHA (prefix)	status
coarse_sweep_sf2.py (c1)	c74c35bc...	unchanged
fine_fit_sf2_v2.py (c3)	dc03007365aa29be...	unchanged
family2_stem_sampled_spike.py (c5)	000c3ef68042f2da6971...	unchanged
family2_stem_sampled_ builder.py (c6)	eea8fb6cb513f342ff71...	unchanged
replay.py post-c11	1f43027039c45f5e066c...	unchanged after cycle 11
stems_6s/drums.wav	34492c03f301b6eac3a7...	unchanged (5 292 044 B)
bass_v2.json and its replay proof	UUID d62cd3b6..., replay SHA 832868d0...aeac5	unchanged; re-reproduced from fresh subprocess at cycle 12

anchor	SHA (prefix)	status
bass.json, bass_family2_v1.json, both bass verdicts, bass arc closeout	—	unchanged
FluidR3_GM.sf2	74594e8f4250680adf59...	unchanged

A.5 Session references

cycle	role	session UUID
10	researcher	67b28d09-ab6b-4c0d-9124-7bc40b5f8b50
10	worker	a46ecebe-098f-47fc-9300-19b34d7f7efe
10	auditor	64ee51ec-9d3b-4512-8f48-0da0c0b79bd4
11	researcher	bcd69ec4-e2f0-4ecf-a989-685050638d1d
11	worker	e90ba69e-0719-4c6c-ae14-4fcd1c07b60
11	auditor	08439cb4-82d7-4539-9f85-7e2117eb4f54
12	researcher	d07132e4-c307-47c7-a8ee-826ef68184cb
12	worker	1771d189-b272-437c-beab-eae25e55920a
12	auditor	c7ac41d8-ccea-4cde-b324-0abe9dba166d

A.6 Cross-reference map

- cycle-9 disk-check false positive → cycle-10 `_disk_ok()` rewrite
 - `test_coarse_sweep_disk_check.py` → cycle-10 detached `coarse_sweep_sf2_drums.py` launch → `drums_sweep_stage1/`
- cycle-10 stage-1 leaderboard → cycle-11 `fine_fit_sf2_drums.py` → `drums_sweep_stage2/leaderboard.tsv` → `drums.json`
- cycle-10 auditor moderate (channel-0-only replay coverage) → cycle-11 `replay.py` L79-93 patch → cycle-11 `drums.replay_proof.json` (with embedded bass-v2 regression block)
- cycle-11 auditor moderate (independent re-verify) → cycle-12 `_replay_regression_c12.py` → `_replay_regression_c12.json` verdict `REPLAY_REGRESSION_HOLDS` → cycle-12 auditor independent re-run reproduces both anchors
- cycle-11 `drums_family_verdict.json` (`SF2_RULED_OUT`) + cycle-12 `drums_family2_verdict.json` (`FAMILY2_RULED_OUT`) → cycle-12 `drums_arc_closeout.json` (`CG_DRUMS_ARC_EXHAUSTED_NO_CONFIRMED`) → cycle-12 `_manager_M-V4-SHOWCASE-1-cg-drums-acceptance-policy.json` (three named options, operator authority) → cycle-13 acceptance-policy resolution
- cycle-7 bass arc closeout shape → cycle-12 drums arc closeout shape (parallel structure); cycle-9 operator directive scope (“Chicken Grease bass”) → cycle-12 escalation refuses unilateral scope extension